

MLE solution

$$\ell_n(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|Y - X\beta\|^2.$$

Constraint:

$$A\beta = c \quad \text{where } A \text{ is an } r \times (p+1) \text{ matrix with full row rank.}$$

Lagrangian:

$$g(\beta, \sigma^2, \lambda) = \ell_n(\beta, \sigma^2) - \lambda^\top (A\beta - c), \quad \lambda \in \mathbb{R}^r.$$

$$\frac{\partial g}{\partial \beta} = 0, \quad \frac{\partial g}{\partial \sigma^2} = 0, \quad \frac{\partial g}{\partial \lambda} = 0.$$

$\Rightarrow$ minimizing $\|X\hat{\beta}_{\text{OLS}} - X\beta\|$ w.r.t. β subject to $A\beta = c$.

Example:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i, \quad \beta_1 = 2\beta_2 + 1.$$

$$\frac{\partial g(\beta, \sigma^2, \lambda)}{\partial \beta} = -2X^\top Y + 2X^\top X\beta - A^\top \lambda = 0.$$

$$\Rightarrow \beta = (X^\top X)^{-1} \left[X^\top Y + \frac{\sigma^2}{2} A^\top \lambda \right] \quad \text{--- (1)}$$

$$\frac{\partial g(\beta, \sigma^2, \lambda)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \|Y - X\beta\|^2 = 0 \quad \Rightarrow \quad \sigma^2 = \frac{\|Y - X\beta\|^2}{n} \quad \text{--- (2)}$$

$$\frac{\partial g(\beta, \sigma^2, \lambda)}{\partial \lambda} = A\beta - c = 0 \quad \Rightarrow \quad A\beta = c \quad \text{--- (3)}$$

From (1):

$$\beta = \hat{\beta}_{\text{OLS}} + \frac{\sigma^2}{2} (X^\top X)^{-1} A^\top \lambda \quad \text{--- (1')}$$

$$A\beta = c = A\hat{\beta}_{\text{OLS}} + \frac{\sigma^2}{2} A(X^\top X)^{-1} A^\top \lambda.$$

As A has full row rank, $A(X^\top X)^{-1} A^\top$ is invertible.

$$[A(X^\top X)^{-1} A^\top]^{-1} (c - A\hat{\beta}_{\text{OLS}}) = \frac{\sigma^2}{2} \lambda \quad \text{--- (4)}$$

Inserting (4) into (1) we get:

$$\hat{\beta}_H = \hat{\beta}_{OLS} - (X^\top X)^{-1} A^\top [A(X^\top X)^{-1} A^\top]^{-1} (A\hat{\beta}_{OLS} - c) \quad \text{--- (5)}$$

Here $\hat{\beta}_H$ signifies the constrained solution of β . (Verify the sign.)

$$\hat{\sigma}_H^2 = \frac{\|Y - X\hat{\beta}_H\|^2}{n} \quad [\text{from (2)}].$$

We need to show that $(\hat{\beta}_H, \hat{\sigma}_H^2)$ maximizes the likelihood.

Will show that for any β satisfying $A\beta = c$

$$\|X\hat{\beta}_{OLS} - X\beta\|^2 \geq \|X\hat{\beta}_{OLS} - X\hat{\beta}_H\|^2 \quad \text{--- (*)}$$

$$\Leftrightarrow \|Y - X\beta\|^2 \geq \|Y - X\hat{\beta}_{OLS}\|^2 + \|X\hat{\beta}_{OLS} - X\hat{\beta}_H\|^2.$$

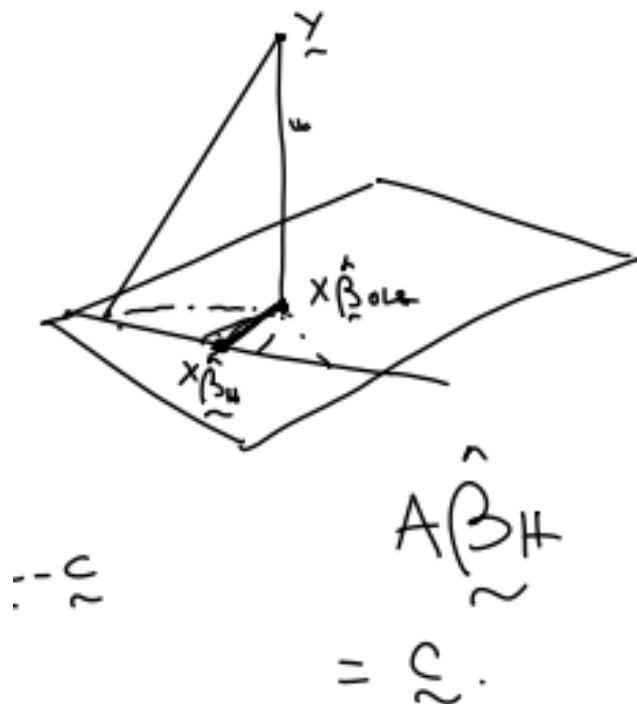
To show (+):

$$\begin{aligned} \|X\beta - X\hat{\beta}_{OLS}\|^2 &= \|X\beta - X\hat{\beta}_H + X\hat{\beta}_H - X\hat{\beta}_{OLS}\|^2 \\ &= \|X\beta - X\hat{\beta}_H\|^2 + \|X\hat{\beta}_H - X\hat{\beta}_{OLS}\|^2. \end{aligned}$$

Product term:

$$\begin{aligned} (\hat{\beta}_{OLS} - \hat{\beta}_H)^\top X^\top X (\hat{\beta}_H - \beta) &= (A\hat{\beta}_{OLS} - c)^\top [A(X^\top X)^{-1} A^\top]^{-1} A(\hat{\beta}_H - \beta) \\ &= 0 \end{aligned}$$

(since $A\hat{\beta}_H - A\beta = c - c$).


$$\|X\beta - X\hat{\beta}_{\text{OLS}}\|^2 \geq \|X\hat{\beta}_{\text{OLS}} - X\hat{\beta}_H\|^2,$$

$\Rightarrow \beta = \hat{\beta}_H$ as $X^\top X$ is invertible. Given the constraint $A\beta = c$,

$$\ell(\beta, \sigma^2) \leq \ell(\hat{\beta}_H, \sigma^2).$$

Next, verify that $\ell(\hat{\beta}_H, \sigma^2) \leq \ell(\hat{\beta}_H, \hat{\sigma}_H^2)$.

Testing of Hypothesis

Consider the normal linear regression model

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2 I).$$

Want to test

$$H_0 : A\beta = c \quad \text{vs} \quad H_1 : \text{no constraint.}$$

LRT

$L^*(\cdot, \cdot)$ denotes the likelihood.

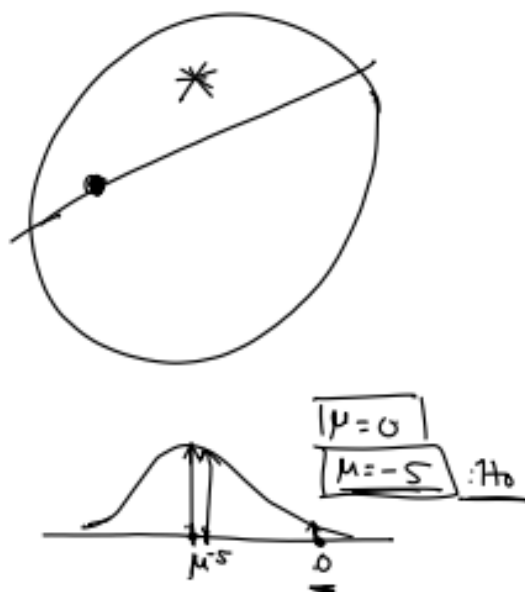
$\ell^*(\hat{\beta}_{H_0}, \hat{\sigma}_{H_0}^2) = \text{maximized log-likelihood(likelihood) under } H_0,$

$\ell^*(\hat{\beta}, \hat{\sigma}^2) = \text{maximized log-likelihood(likelihood) without any constraint.}$

If

$$\Lambda(y) = \frac{L^*(\hat{\beta}_{H_0}, \hat{\sigma}_{H_0}^2)}{L^*(\hat{\beta}, \hat{\sigma}^2)}$$

is small then we reject H_0 . (Equivalently, $\ell^*(\hat{\beta}_{H_0}, \hat{\sigma}_{H_0}^2) - \ell^*(\hat{\beta}, \hat{\sigma}^2)$.)



Under H_0 :

$$\begin{aligned}\hat{\beta}_{H_0} &= \hat{\beta}_{OLS} - (X^\top X)^{-1} A^\top [A(X^\top X)^{-1} A^\top]^{-1} (A\hat{\beta}_{OLS} - c), \\ \hat{\sigma}_{H_0}^2 &= \frac{\|Y - X\hat{\beta}_{H_0}\|^2}{n}.\end{aligned}$$

$$\begin{aligned}\ell^*(\hat{\beta}_{H_0}, \hat{\sigma}_{H_0}^2) &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \hat{\sigma}_H^2 - \frac{1}{2\hat{\sigma}_H^2} \|Y - X\hat{\beta}_H\|^2 \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \hat{\sigma}_H^2 - \frac{n}{2}.\end{aligned}$$

Unconstrained:

$$\begin{aligned}\hat{\beta} &= \hat{\beta}_{MLE} = (X^\top X)^{-1} X^\top Y, \\ \hat{\sigma}^2 &= \hat{\sigma}_{MLE}^2 = \frac{\|Y - X\hat{\beta}_{MLE}\|^2}{n}.\end{aligned}$$

$$\ell^*(\hat{\beta}, \hat{\sigma}^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \hat{\sigma}_{MLE}^2 - \frac{n}{2}.$$

Ratio of likelihood:

$$\Lambda(y) = \left(\frac{\hat{\sigma}_H^2}{\hat{\sigma}^2} \right)^{-n/2}.$$

$$\begin{aligned}\frac{\hat{\sigma}_H^2}{\hat{\sigma}^2} &= \frac{\|Y - X\hat{\beta}_H\|^2}{\|Y - X\hat{\beta}_{OLS}\|^2} = \frac{\|Y - X\hat{\beta}_{OLS} + X\hat{\beta}_{OLS} - X\hat{\beta}_H\|^2}{\|Y - X\hat{\beta}_{OLS}\|^2} \\ &= \frac{\|Y - X\hat{\beta}_{OLS}\|^2 + \|X\hat{\beta}_{OLS} - X\hat{\beta}_H\|^2}{\|Y - X\hat{\beta}_{OLS}\|^2} = 1 + \frac{\|X\hat{\beta}_{OLS} - X\hat{\beta}_H\|^2}{\|Y - X\hat{\beta}_{OLS}\|^2}.\end{aligned}$$

Denominator:

$$\frac{RSS}{\sigma^2} = \frac{\|Y - X\hat{\beta}_{OLS}\|^2}{\sigma^2} = \frac{Y^\top (I - P(X)) Y}{\sigma^2} \sim \chi_{n-p-1}^2.$$

$$\frac{\chi_n^2/n}{\chi_n^2/m} \sim F_{n,m}, \text{ if both the chi squares are independent}$$

Numerator:

$$\begin{aligned}\|X(\hat{\beta}_H - \hat{\beta}_{OLS})\|^2 &= (A\hat{\beta}_{OLS} - c)^\top [A(X^\top X)^{-1}A^\top]^{-1} A(X^\top X)^{-1}A^\top [A(X^\top X)^{-1}A^\top]^{-1} (A\hat{\beta}_{OLS} - c) \\ &= (A\hat{\beta}_{OLS} - c)^\top [A(X^\top X)^{-1}A^\top]^{-1} (A\hat{\beta}_{OLS} - c).\end{aligned}$$

$= \sigma^2 \tilde{Z}^\top \tilde{Z}$ where

$$\begin{aligned}A\hat{\beta}_{OLS} &= A(X^\top X)^{-1}X^\top Y \sim N(A\beta = c, \sigma^2 A(X^\top X)^{-1}A^\top), \text{ under } H_0 \\ (A\hat{\beta}_{OLS} - c) &\sim N(0, \sigma^2 A(X^\top X)^{-1}A^\top), \text{ under } H_0\end{aligned}$$

$$\tilde{Z} = \frac{1}{\sigma} [A(X^\top X)^{-1}A^\top]^{-1/2} (A\hat{\beta}_{OLS} - c) \sim N(0, I).$$

$$\begin{aligned}\Sigma &> 0, \quad \Sigma = UDU^\top, \quad D = \text{Diag}(d_1, \dots, d_r), \\ d_j &> 0 \quad \forall j, \quad U^\top U = I, \\ \Sigma^{1/2} &= UD^{1/2}U^\top, \quad D^{1/2} = \text{Diag}(\sqrt{d_1}, \dots, \sqrt{d_r}), \\ \Sigma^{-1/2} &= UD^{-1/2}U^\top, \quad D^{-1/2} = \text{Diag}(1/\sqrt{d_1}, \dots, 1/\sqrt{d_r}), \\ \Sigma^{-1/2}\Sigma^{1/2} &= \Sigma^{-1}.\end{aligned}$$

So

$$\tilde{Z}^\top \tilde{Z} \sim \chi_{(r)}^2.$$

$$\frac{\|X\hat{\beta}_{OLS} - X\hat{\beta}_H\|^2}{\sigma^2} \sim \chi_r^2.$$

Observe that $\|X\hat{\beta}_{OLS} - X\hat{\beta}_H\|^2$ is a function of $\hat{\beta}_{OLS}$ only, and $\hat{\beta}_{OLS} \perp \text{RSS}$.

So the numerator and denominator are independent.